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Impact of poly naphthalene sulfonate on the dispersion stability of iron oxide nanoparticles

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ABSTRACT

Iron oxide nanoparticles (IOXn) are unstable in aqueous solutions, which prevents their use in various applications. Because of this, it was proposed to treat the IOXn surface by employing a dispersant like sodium poly (naphthalene formaldehyde) sulfonate (PNS). In a single easy process, the (IPNS) were made by covering the IOXn with PNS. The material's structure, morphology, and magnetic properties were identified using XRD, TEM, Zeta potential, FTIR, HR-SEM, and a VSM. Due to its complex molecular structure and high anionic charge density, the polymeric dispersant (PNS) displays various characteristics. PNS successfully modifies the IOXn surface to create excellent colloidal electrostatic stability. At pH = 8, the zeta potential decreased from −13 to −23 mv. After the coating procedure, the crystalline structure changed into an amorphous one. The average particle size was decreased from 138 to 32 nm. In addition, there was a slight reduction in the magnetization saturation (Ms).

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1. Introduction

Low grain size magnetite nanoparticles (MgN) display superparamagnetic activity, although at ambient temperature, their coercivity and remanence values are low. The colloidal dispersion of MgN is improved with coating or capping agents, such as oleic acid, polyethylene glycol [1], dextran [2], or encapsulated in polystyrene to increase its performance [3]. There are many potential uses for the MgN, including removing heavy metals. [4], drug delivery agents [5], energy storage [6], and cancer therapy [7].

The control of size, clustering, and aggregation of MgN is a significant challenge during the synthesis, handling, and applications. MgN has high surface energies related to volume ratio; it aggregates to reduce the high surface energy [8]. The anionic surfactant is absorbed on the surface of nanoparticles which inhibits the accumulation and modifies the colloidal stability in the solution. Through co-precipitation, time and temperature are affected in the colloidal stability and size distribution (non-uniform shape) of nanoparticles [9]. The phase composition of magnetite nanoparticles in an air environment depends on the initial molar ratio Fe^{2+} : Fe^{3+} . The molar ratio of Fe^{2+} : Fe^{3+} changes as Fe^{2+} oxidizes to Fe^{3+} . Maghemite replaces magnetite as a result of the phase composition

change [10]. A synthetic dispersant with good wetting, emulsifying, and foaming properties is PNS condensate. It is frequently used as a dispersion in coatings, dye dispersant, and reducer of water in cement (reducing agent), dyestuff industry, flotation separation, and coal water slurry [11–14]. PNS was created by combining concentrated sulfuric acid and formaldehyde with naphthalene. Condensates with longer chains exhibit a stronger diffusion impact than those with shorter chains. PNS has adsorption behaviors for components such as carbonaceous material, kaolinite, quartz, and pyrite on their particle surfaces [15]. The dispersion mechanism depends on the hydrophobic tails (naphthalene) and the hydrophilic groups (sulphonated group) in PNS [16]. (IPNS) has not yet been reported on for dispersion stability. In contrast to current procedures, the study's goal is to quickly and cheaply produce coated IOXn with a high magnetic value in an aqueous solution.

2. Experiment

2.1. Materials

Ferrous sulfate $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, Ferric chloride $\text{FeCl}_3 \cdot 7\text{H}_2\text{O}$, sodium hydroxide pellets solid, formaldehyde, sulphuric acid, and Ethyl alcohols were purchased from Piochem Co. India. Naphthalene (99 %) from Merck Sigma- Aldrich.

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2.2. Characterizations

2.2.1. Fourier transforms infrared spectroscopy analysis (FT-IR)

One milligramme of each sample of raw and modified alumina silicate was ground and specified in the range of $4000\text{--}400\text{ cm}^{-1}$ with a resolution of 2 cm^{-1} on the spectrometer model type Mattson-Infinity Series Benchtop 961 that was used.

2.2.2. Particle size and zeta potential

The particle size distributions and zeta potential were measured with Malvern Zeta sizer nano series (NANO-ZS) HT using dynamic light scattering in ethanol.

2.2.3. Wide angle X-ray diffraction.

WAXD were measured using a Pan Analytical Model X Pert Pro with CuK α radiation ($\lambda = 0.1542\text{ nm}$), a Ni-filter, and a general area detector. The diffractograms were measured over a two-hour period at temperatures ranging from 10 to 40 °C, with a step size of 0.02 °A and a step time of 0.7000 s with generator settings of 40 MA and 40 kV.

2.2.4. Vibrating sample magnetometer (VSM)

Magnetic measurements were executed with the Quantum Design Model 6000 and parameters such as (Ms), (Hc) and (Mr) were evaluated.

2.2.5. Thermogravimetric (TGA) and differential thermal (DTA) analysis

On the SDTQ-600 (TA-UAS) thermobalance instrument, 15 mg of the sample was heated above 600 °C at a rate of 10 °C min^{-1} in a nitrogen flow at a rate of 100 mL min^{-1} .

2.2.6. Surface area analysis

Quantachrome NOVA Automated Gas Sorption was used to determine the surface area. Pure liquid N₂ adsorption at 200 °C for 2 h was used to measure the samples, with P/P₀ values ranging from 0.05 to 1.0.

2.2.7. Gel permeation chromatograph (GPC)

The average molecular weights (M_w and M_n) of sodium poly (naphthalene formaldehyde) sulfonate were determined by Waters Model 515/2410 using a gel Permeation Chromatographic technique (GPC, Waters, America).

2.2.8. Contact angel

The contact angles were tested on the glass surface. The contact angles were recorded for 15 s.

2.2.9. Ultraviolet–visible (UV–vis) absorption spectroscopy

UV–vis absorption measurements of the magnetite, PNS, and IPNS in aqueous solution were detected by using Jenway 6505 UV/vis spectrometer (UK).

2.2.10. Field emission scanning electron microscope (FE-SEM)

The surface morphology was detected by using ZEISS (FEI Quanta FEG 250).

2.2.11. Transmission electron microscope (TEM)

Copper grids with a very thin amorphous carbon layer were used to study TEM photos. The carbon-coated grid was used to deposit the prepared aqueous IPNS. The size of nanoparticles was determined using a JEOL JEM-model that operated at 200 kV and was equipped with a CCD camera.

2.3. Preparation of magnetite nanoparticles

A precipitation process was used to synthesize the magnetite nanoparticles, which involved combining FeCl₃·7H₂O and FeSO₄·7H₂O with molar ratio (1:1) in deionized water. 1 N of NaOH was added to the previous solution until the pH value was set at 8.5. For three hours, the temperature was raised to 90°Celsius. The precipitated nanoparticles were washed four times with deionized water [17].

2.3.1. Synthesis of sodium poly (naphthalene formaldehyde) sulfonate (PNS)

A three-necked flask was filled with 60 g of naphthalene and heated to 137 °C. A magnetic stirrer was used to stir the mixture at the same time. Drop by drop, over the period of thirty minutes, 30 mL of 98 % concentrated sulfuric acid was added to the flask (the stirrer speed was too slow at low temperature). The reaction was maintained at this temperature for 180 min after the temperature was raised to approximately 160 °C. The temperature was then decreased to around 140 °C, and 20 min were spent hydrolyzing with 10 mL distilled water. When the temperature reaches 110 °C, add 20 mL formaldehyde, leave the reactor at that temperature for 5 h, and allow it to cool after the reaction, then add sodium hydroxide to bring the pH level down to 7.8. [15].

2.3.2. Preparation dispersion solution of IPNS

PNS was prepared at concentrations of 0.1, 0.2, and 0.3 M. First, the magnetite powder (0.1 gm per liter) was dispersed in distilled water via ultrasonication. Next, PNS solution was added to magnetite in the nano-form. Then the mixture was dispersed again via ultrasonication.

3. Results and discussion

3.1. Gel permeation chromatography

Sodium poly (β - naphthalene formaldehyde) sulfonate (PNS) was synthesis through 4 steps. The first step is the sulfonation process by adding a sulfonic group ($-\text{SO}_3\text{H}$). The sulfonation group is encouraged to go towards the β - position by the thermodynamic reaction at elevated temperature. It is possible to hydrolyze the α -naphthalene sulfonic acid (second step) at an appropriate temperature. The stability of β -naphthalene sulfonic acid increases with temperature over 120 °C. Therefore, a high molecular weight polymer is created in the third stage by condensing β -naphthalene sulfonic acid with formaldehyde at 110 °C. Alkali neutralizes excess sulfuric acid and alkyl sulfonic acid produced by the reaction into salts during the final step of production [15]. GPC shows average molecular weight, Polydispersity (PDI), and the number and weight average molecular weights of the prepared PNS. The average number molecular weight (M_n) is 12381; average weight molecular weight (M_w) is 26,561 Dalton, and polydispersity index (M_w/M_n) is 2.145. The M_n and M_w are not nearly equal values. PDI expresses molecular weight distribution, the high PDI value indicated a non-uniform distribution with significant molecular length diversity. This means that not all molecules are treated the same way. PNS's lengthy chains produce steric forces, which have a greater dispersive effect than electrostatic forces.

3.2. FT-IR analysis

FTIR spectrum of PNS is shown in Fig. 1. The spectrum shows two bands at 3850 and 3425 cm^{-1} corresponding to the stretching vibrations of the hydroxyl group's free and hydrogen bond, respectively; the bands at 2945 and 2881 cm^{-1} are related to the C–H

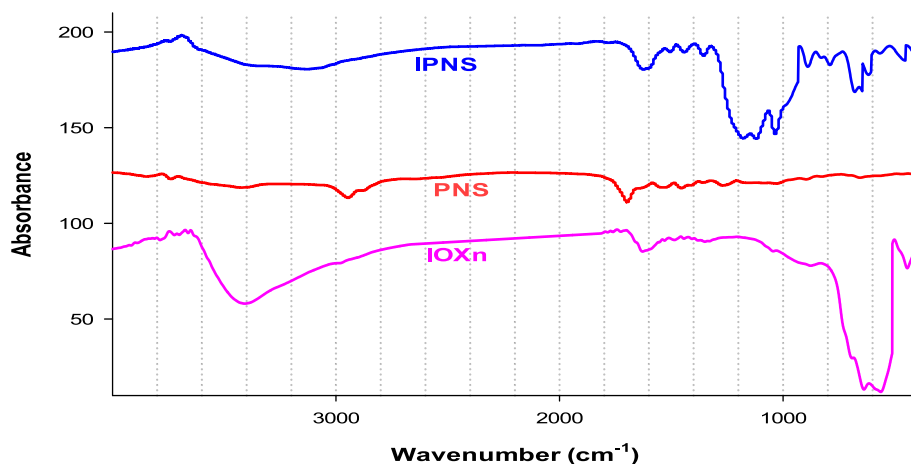


Fig. 1. FTIR spectra of IOXn, PNS, and IPNS.

group. Vibrations at 1688 cm^{-1} identify the aromatic rings, while the CH_2 bridging group emerges at 1455 cm^{-1} . The sulfate ions ($\text{S}=\text{O}$ group) emerge symmetrically stretching vibration at 1030 cm^{-1} , whereas the alkyl sulfonate ions ($\text{S}=\text{O}$ group) appear as weak peaks at 1351 and 1163 cm^{-1} , respectively. Two bands are attributed to a substituted ring structure that arises at $893\text{--}830\text{ cm}^{-1}$ [18–20].

Fig. 1 also shows the FTIR spectrum of IOXn. Two absorption bands in the $638\text{--}563\text{ cm}^{-1}$ range were formed by dividing the band by over 570 cm^{-1} , and these bands were attributed to the vibration of iron oxide bonds at tetrahedral and octahedral sites. At about 440 cm^{-1} , was transferred from the band at 370 cm^{-1} and given sole responsibility for the iron octahedron. It is equivalent to bulk iron oxide's Fe-O. The bandwidth at 3142 cm^{-1} was attributed to O–H stretching vibration. OH group or water molecules were blamed for the band at 1628 cm^{-1} [21].

FTIR spectrum of IPNS is shown in Fig. 1. The vibrating band at 1688 cm^{-1} in PNS disappeared, and the peak at 1626 cm^{-1} appeared in IPNS spectrum. Broadband at 3142 cm^{-1} is still exists. Three strong peaks were observed at 1178 , 1122 , and 1034 cm^{-1} ; both symmetric stretching vibration and asymmetric stretching vibration were given to them. ($-\text{SO}_3^-$), respectively. Two absorption bands in the $638\text{--}563\text{ cm}^{-1}$ region were not changed; it is attributed that iron oxide is not oxidized due to the coating process. The dispersion mechanism of anionic polymers (PNS) depends on a net negative electrical charge ($-\text{SO}_3^-$) to the surface of the magnetite nanoparticles; this causes repelling forces to form between neighboring particles and protects them from settling for a long time.

3.3. Wide-angle X-ray diffraction (WAXD) analysis

As shown in Fig. 2, X-ray diffraction investigations were conducted to investigate the amorphous and crystalline character of the produced IPNS in comparison to IOXn. When compared to those at (220), (311), (400), (422), (511), and (440), the primary distinct peaks of the crystallographic plane for pure iron oxide were assigned at (30.20°), (35.58°), (43.2°), (53.5°), (57°), and (62.7°). The diffraction peaks' widening reveals the size of the magnetite nanoparticles. The XRD patterns of IPNS show the main characteristic diffraction peaks (2θ) at (20.99°), (29.91°), (31.72°), (33.59°), (35.29°), (36.41°), (40.87°), (53.12°), (56.91°), (58.73°), and (61.13°), it has amorphous structures. The magnetite's primary distinctive peaks vanished. Due to the partial oxidation of magnetite at the surface, new peaks have emerged; however, the bulk has not undergone oxidation. When acid and iron oxide nanoparti-

cles were mixed and dried, a minor amount of maghemite ($\gamma\text{-Fe}_2\text{O}_3$) was created by oxidation; its distinctive peaks were at (2θ) 21.18° , 33.19° , and 35.6° [22]. The XRD pattern of IPNS explained that the IOXn were coated with PNS on their surfaces. So the crystalline structure of magnetite was coated with new amorphous structures. The size of magnetite particles and IPNS (solid powder) have been estimated using X-ray diffraction from the Scherrer formula Eq. (1):

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (1)$$

Where ' λ ' is the wavelength of X-ray, ' β ' is full width at half maximum was measured in radians, ' θ ' is the diffraction angle and ' D ' is particle size average diameter.

The IOXn and IPNS were calculated to have crystals with diameters of 164 and 6 nm, respectively. The obtained results provide an explanation for the increase in particle size distribution values.

3.4. Vibrating sample magnetometer (VSM)

Magnetic characteristics are crucial in a variety of applications. Fig. 3 depicts the IOXn and IPNS hysteresis loop. At room temperature, the Magnetization Saturation (M_s) of IOXn and IPNS were (62) and (37.8) emu/g, respectively. IOXn have superparamagnetic characteristics, according to the findings. The huge particle size of IOXn is responsible for its high Magnetic Saturation value. As a result, the M_s of magnetite nanoparticles diminishes as their particle size decreases. In the general case of iron oxide nanoparticles, reducing the size of single particles to the nanoscale (below 20 nm) significantly impacts the magnetic ordering of surface spins. In addition, due to PNS adsorption on the surface of magnetite, M_s for IPNS is lower than M_s for pure IOXn. The difference between IOXn and IPNS readings is negligible. The coercive force and magnetic remanence of the IPNS are also minimal. As a result, the magnetite and IPNS prepared are highly attracted to magnetic fields. Iron oxide nanoparticles alone exhibit surface spin disorder growth and saturation magnetization decrease due to the smaller particle size increasing the surface-to-volume ratio [23]. The results show that the particle size of the IPNS is less than IOXn, but the M_s value was decreased due to the coating process.

3.5. Zeta potential analysis and dynamic light scattering analysis (DLS)

The stability and dispersibility of particles in a solution are governed by their surface charges. The precipitation process is influenced by the particle size [24]. HCl (0.1 M) and NaOH (0.1 M)

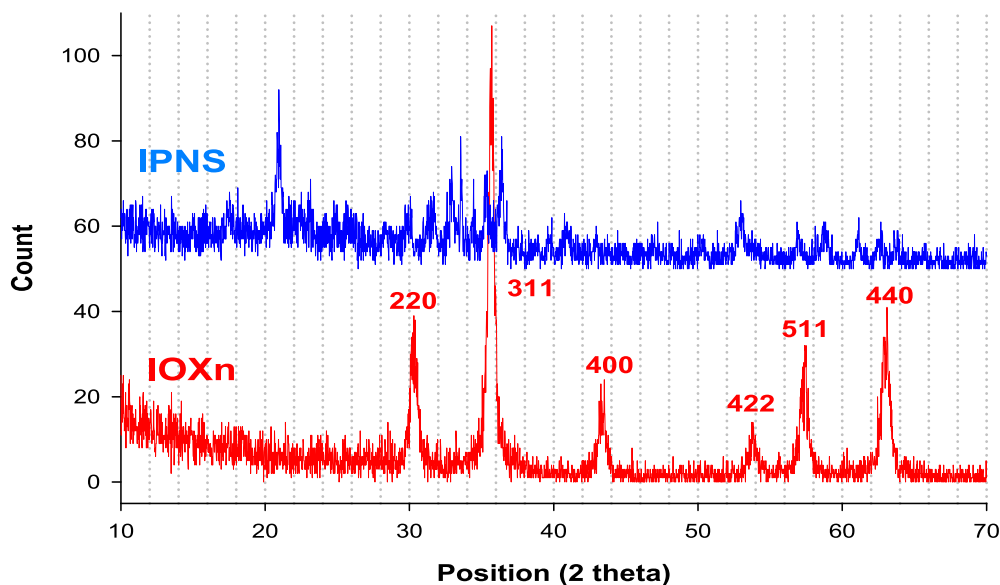


Fig. 2. WXR D patterns of IOXn, and IPNS.

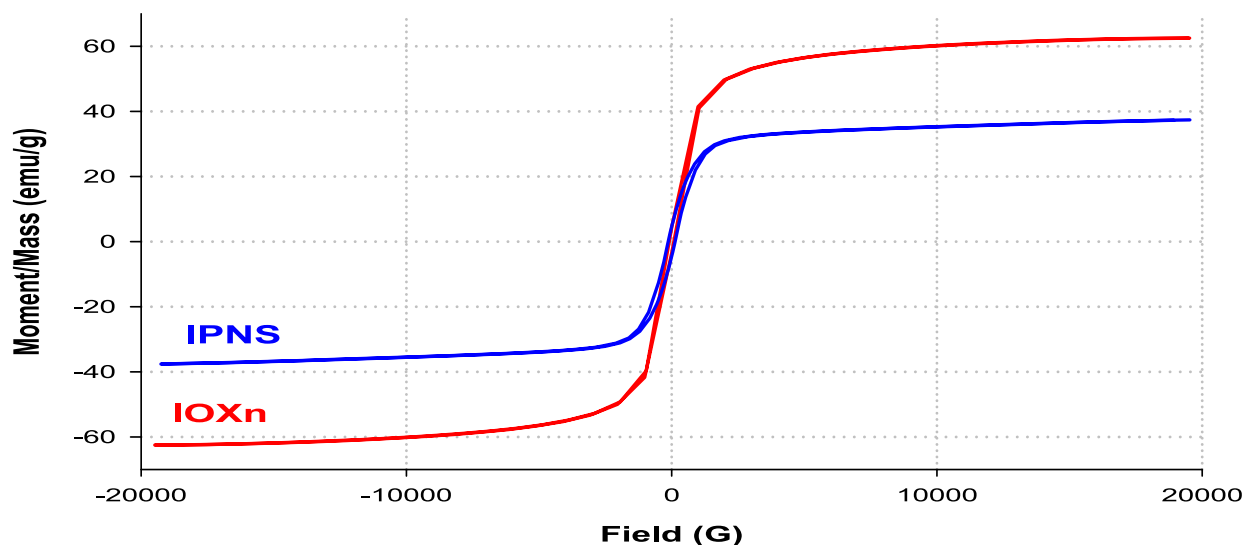


Fig. 3. Hysteresis loops of IOXn, and IPNS.

were used to adjust the pH from 2 to 11 of dispersed IPNS. During the pH correction, the magnetic particle dispersion was circulated. Fig. 4a and b shows the zeta potential (at pH = 8) and particle size of pure IOXn, which are -13 mV and 138 nm, respectively. PNS can be ionized negatively, resulting in a hydrophilic surface on the particle's surface, which might affect the particle's zeta potential values. Fig. 4c illustrates how pH affects coated particles' zeta potential and particle size distribution. The Zeta potential of IPNS was changed from a negative value (-80 mV) to a positive value ($+60$ mV) at different pH values because of the adsorption PNS on their surface. Additionally, there is an electrostatic contact between the sulfonated group of PNS and the (OH) alkaline of IOXn. There's also the electrical repulsion between sulfonated groups that coat the IOXn. In acidic media, PNS is only slightly soluble, But it is completely soluble (high zeta potential) in alkaline media. Due to the reverse back to ferrous and ferric chloride, IOXn is soluble in acidic media (high zeta potential) while precipitating in alkaline media. As a result, the zeta potential of IPNS is high at

both high and low pH levels. IPNS particles range in size from 31 to 47 nm. The best average particle size (31 nm) is found at pH = 8 because of the PNS solubility fault in acidic media, while excessive co-precipitation of iron oxide occurred in alkaline media. However, it is high (47 nm) in acidic and alkaline media. At pH = 8, IPNS has higher stability and dispersibility.

3.6. Field emission scanning electron microscopy (FE-SEM)

FE-SEM was used to examine the surface morphology, as shown in Fig. 5. The nanoparticles' physical properties and geometric shape are critical in determining a nanoparticle's efficiency and applicability. The average particle size of IOXn is around 100 nm (before coating); their morphological appearance was irregularly formed with acute and webbed angles. The IOXn are near in size. Furthermore, the particles form agglomerations with one another. The particle size of IPNS decreases and the aggregation of particles decreases. The average particle size of IPNS is 35 nm, with a high

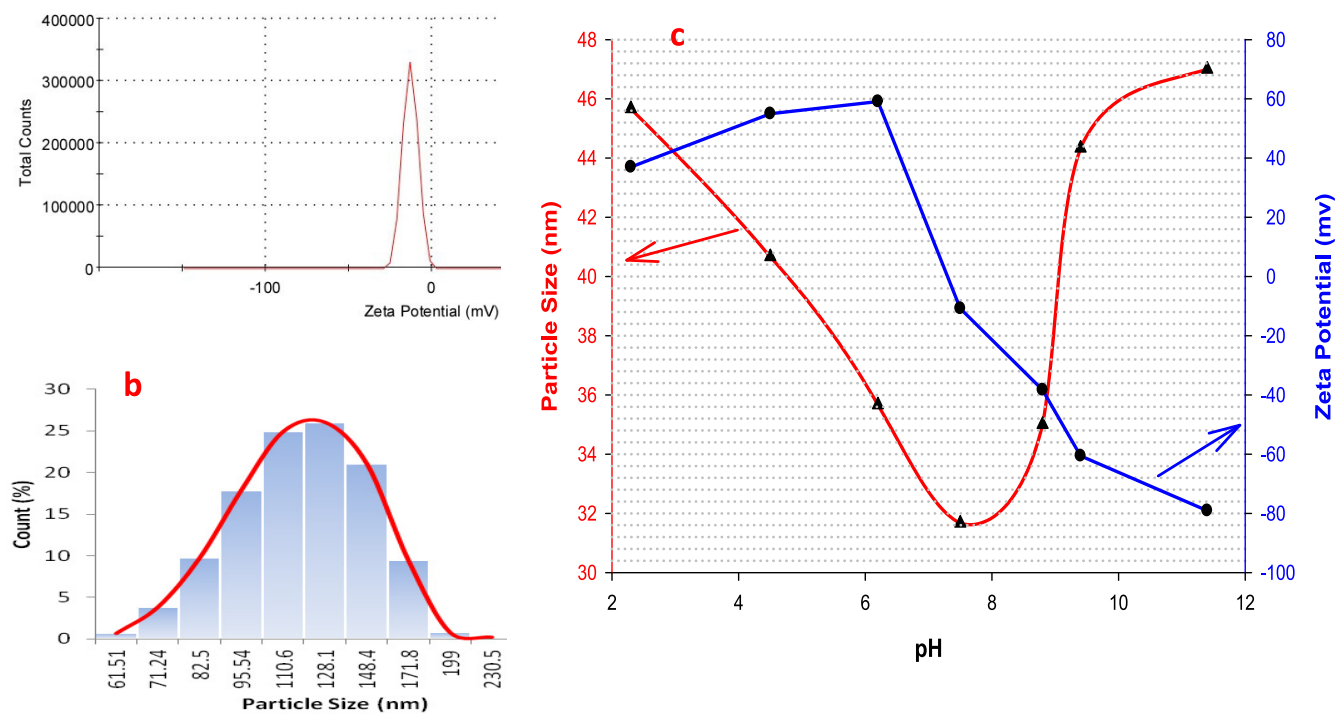


Fig. 4. (a) Zeta potential and (b) Particle size distribution of IOXn and (c) Relation between particle size distribution, pH, and zeta potential of IPNS.

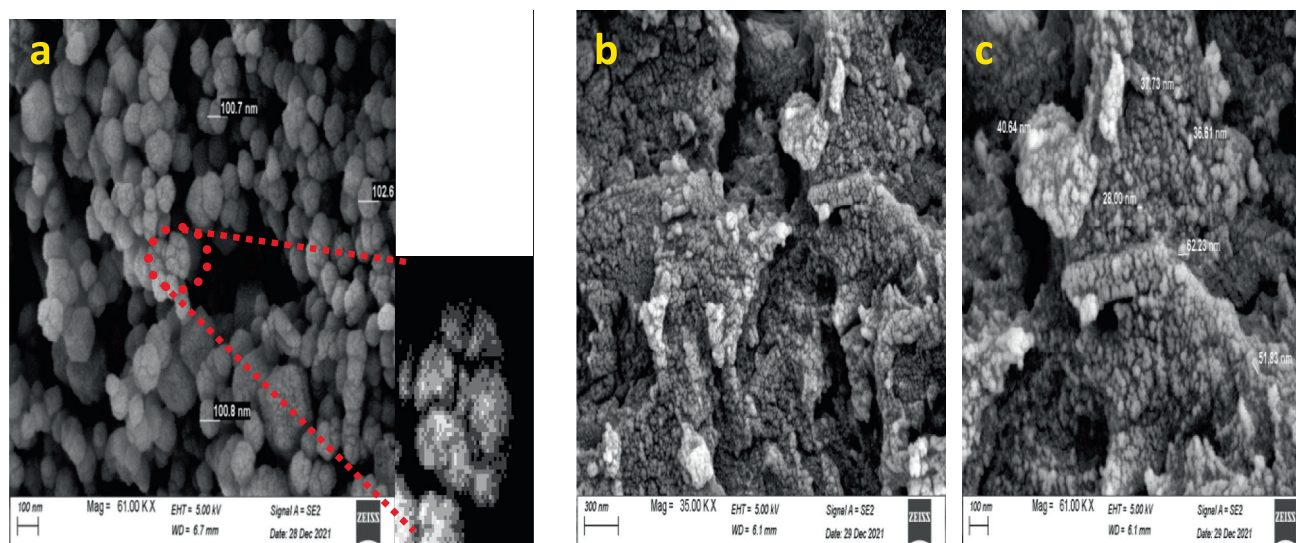


Fig. 5. SEM images of (a) IOXn, (b,c) IPNS with two magnifications.

surface area and porosity. Because the cathode in the electron gun has a high electron energy (of FE-SEM instrument), PNS could partially melt on the surface of iron oxide. The photos depicted a similar shape with uniform particle dispersion. Due of its capacity to provide steric repulsion against an accumulation, the surface of the iron oxide has been successfully coated with PNS, which acts as a protective layer against Vander wall's attraction. The IOXns' surface is permeable to PNS, and the steric interaction between two particles moving toward one another depends on their distance from one another. As a result, the opposing particles' polymer chains are intertwined. As a result, the particles are being

pushed apart. When nanoparticles are functionalized, their properties change.

3.7. Surface area analysis

The specific surface area of magnetite is plotted in Fig. 6, and they were measured from nitrogen adsorption-desorption isotherms. The curve shows type IV isotherm with an H3-type hysteresis loop. Numerous aspects, including as hydroxylation, dissolution, interaction with adsorbate molecules, and stability are influenced by the specific surface area. It is also important to

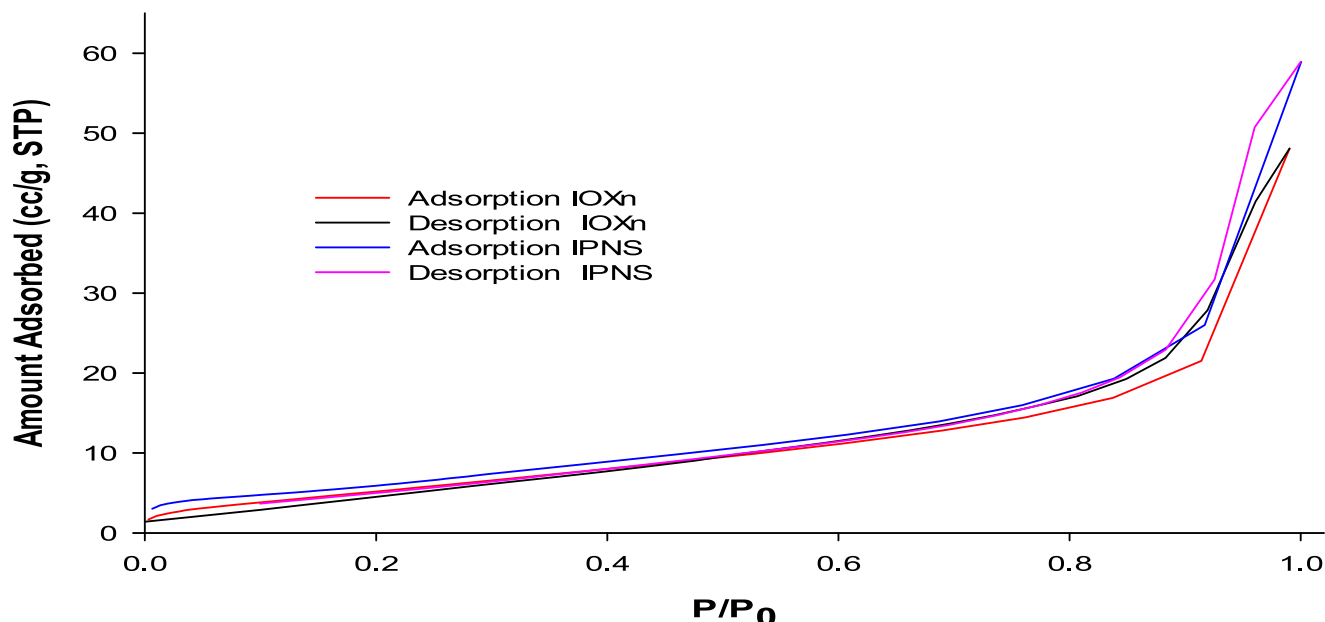


Fig. 6. Nitrogen adsorption-desorption isotherms of IOXn.

note that the technique for material preparation is dependent on the specific area. Magnetite samples have been reported to have a range of 4 to 100 m²/g depending on the size of the crystal, with the bigger crystals being non-porous. It is possible to determine specific surface, exterior, and micropore areas as well as pore diameter, volume, and distribution using data from the isotherms of BET [25]. The total specific surface area of IOXn, and IPNS is 35.049 and 30.686 m²/g; the external specific surface area is 23.866 and 22.218 m²/g, and the pore diameter is 2.208 and 2.806 nm, respectively; they were measured at high partial pressure ($P/P_0 = 1$). The pore diameter was decreased according to the coating process. The N₂ gas was adsorbed on the inner surface of iron oxide; the mesopores structure allows adsorbing a single layer to several layers, according to a little increase in adsorption. The desorption isotherm is above the adsorption isotherm at the higher relative pressure stage. The PNS decreases the adsorption of N₂ gas on the surface of iron oxide nanoparticles, which means the PNS covers the external and internal pores and interlayer space.

3.8. High resolution- transmission electron microscope (HR-TEM)

The morphologies of the produced samples IOXn and IPNS were examined using transmission electron microscopy (TEM), as shown in Fig. 7. The suspensions of iron oxide (uncoated and coated) were prepared and stirred for 15 min with a mechanical agitator. Then 5 μ L of the suspension were poured onto a carbon-coated copper grid. The IOXn appear as aggregated small particles and large hexagonal nanoplatelets of ferrous oxides with a mean diameter of 150 nm, as indicated in the surface morphologies in Fig. 7a [26]. The nanoparticles also didn't appear as a uniform dispersed size. As seen in Fig. 7(b–d), the IPNS surface morphologies explain that the nanoparticles are better disseminated and isolated from one another than the IOXn; their average particle sizes are 40 nm, and they appear as dense color according to coating. The nanoparticle's size did not alter (before and after modification), but the difference was in the degree of aggregation. The PNS dispersion mechanism occurs because the PNS anionic polymers can adsorb on the surface of the IOXn. Therefore the nanoparticles carry a net negative electrical charge, causing repelling interactions

between neighboring and increasing dispersion [24]. SEM and TEM morphologies of IOXn are similar in average particle size and match with DLS data. The SAED pattern has polycrystalline characteristics, implying that the IPNS are oriented randomly. As a result, the building blocks' monodispersity (IOXn) was transformed into polydispersity (IPNS). Because of the PNS coating on the surface of IOXn, the diffracted beam passes through the polycrystalline structures, as seen in Fig. 7e.

3.9. Contact angle (CA) and surface activity

The contact angle was measured at the glass wall's surface. Each test was carried out in a laboratory setting with a constant drop size throughout all tests and a room temperature of 25 °C. The test was measured three times per sample. This measurement was utilized to see how the wettability of PNS affected the magnetite nanoparticles. The contact angle is (43°), (36°), (41°), and (36°) for distilled water, PNS (0.1 M), IOXn (0.1 g/L), and IPNS (pH = 8.5), respectively, as shown in Fig. 8. Due to the anionic polymer's migration from the bulk to the surface, the contact angle of the PNS solution is lower than that of water. Additionally, the contact angle of water is reduced by the presence of IOXn. The hydrophilic characteristics of the IOXn are enhanced and the contact angle is decreased by the presence of PNS on the surface. The surface tension and contact angle tests were measured with the same instrument. The surface tension is (71.48), (71.41), (71.1), and (71.1) at 25 °C, for distilled water, PNS (0.1 M), IOXn (0.1 gm/L), and IPNS (pH = 8.5), respectively. Adding PNS or IOXn does not change water's surface tension. PNS does not create micelle, so it is not considered a surfactant.

3.10. Thermogravimetric (TGA) and differential thermal (DTA)

TGA and DTA for IOXn and IPNS are presented in Fig. 9. At 200 °C, the thermogravimetric curve reveals a weight loss of around 5.96 percent due to residual free water loss from interlayer space. Low molecular weight oligomers were also decomposed in this region. At 400 °C, the partial collapse of the mesopore structure of low molecular weight polymers or burning of any residual hydrocarbon template results in a weight loss of 4.19 percent. Two

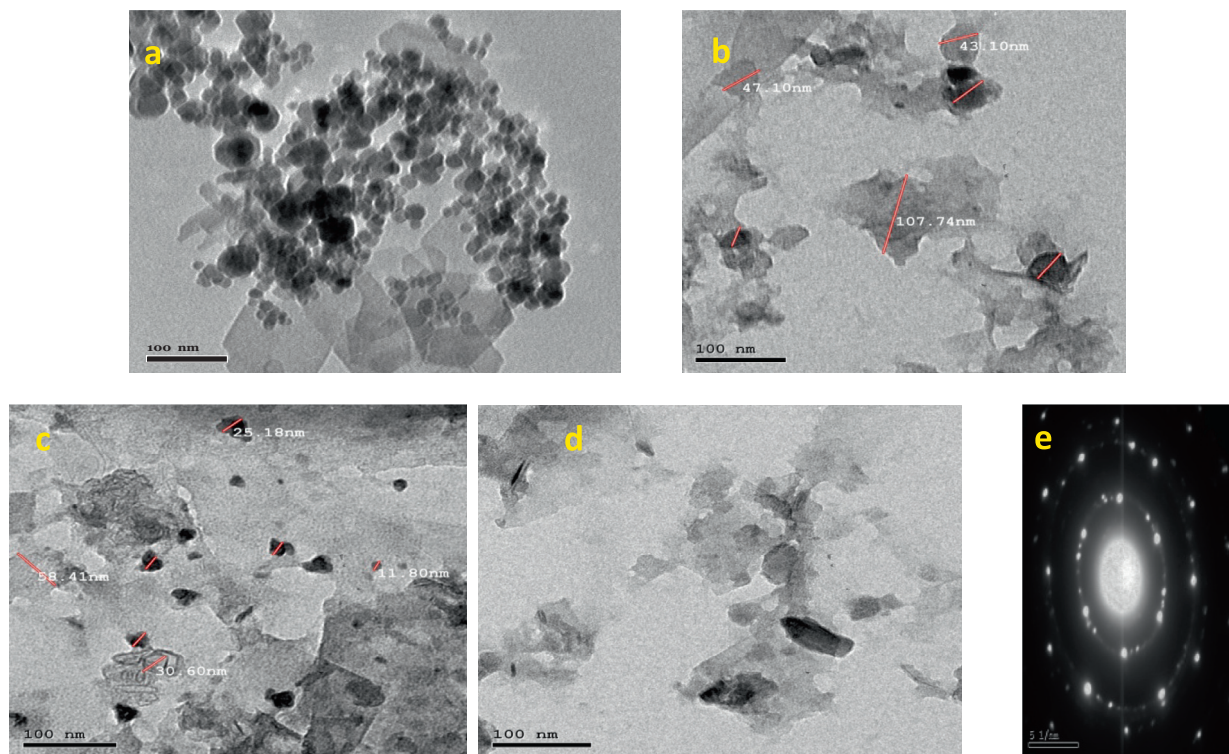


Fig. 7. HRTEM images of (a) IOXn, (b-d) dispersed IPNS and (e) SAED pattern of IPNS.

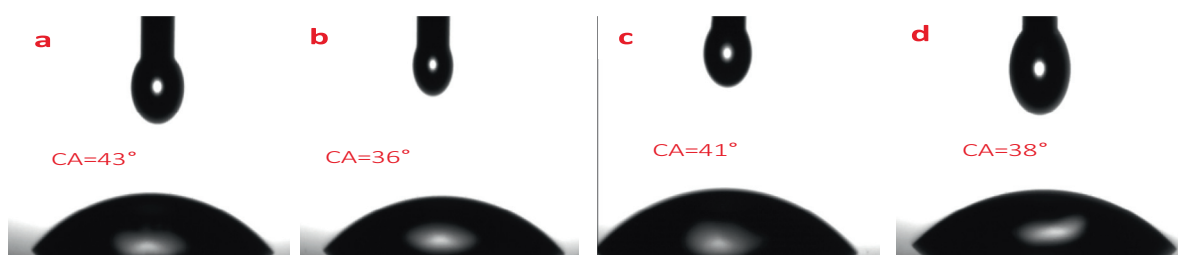


Fig. 8. Contact angle of (a) distilled water, (b) PNS (0.1 M), (c) magnetite (0.1gm/L), and (d) IPNS.

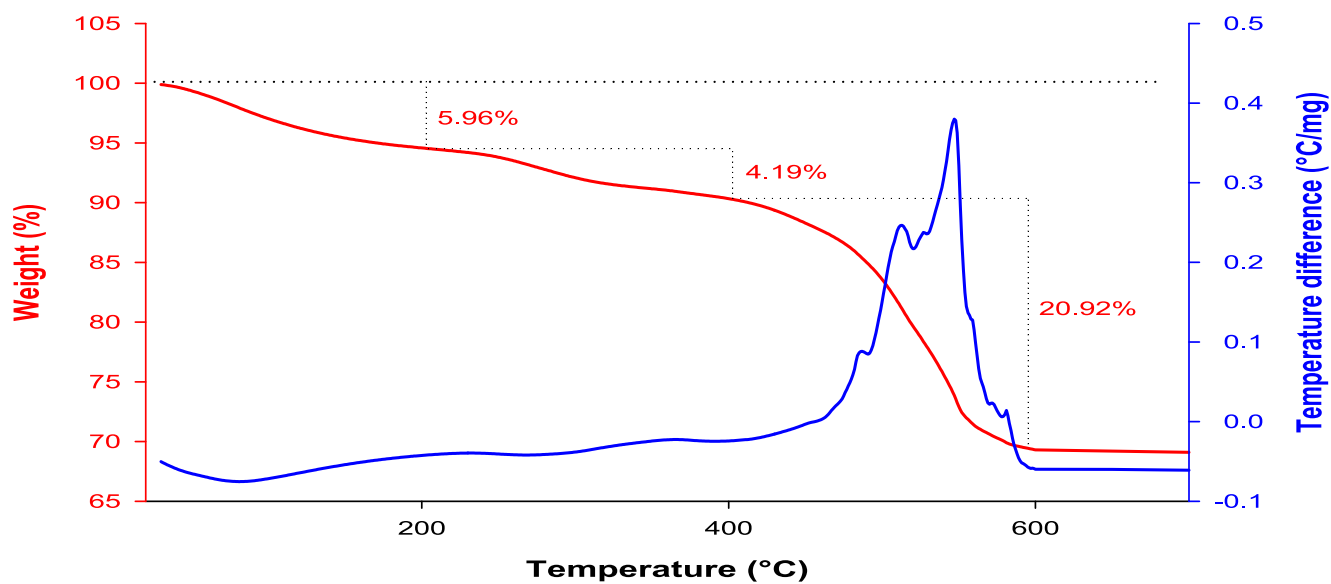


Fig. 9. Thermogravimetric (TGA) and differential thermal (DTA) curve of IPNS.

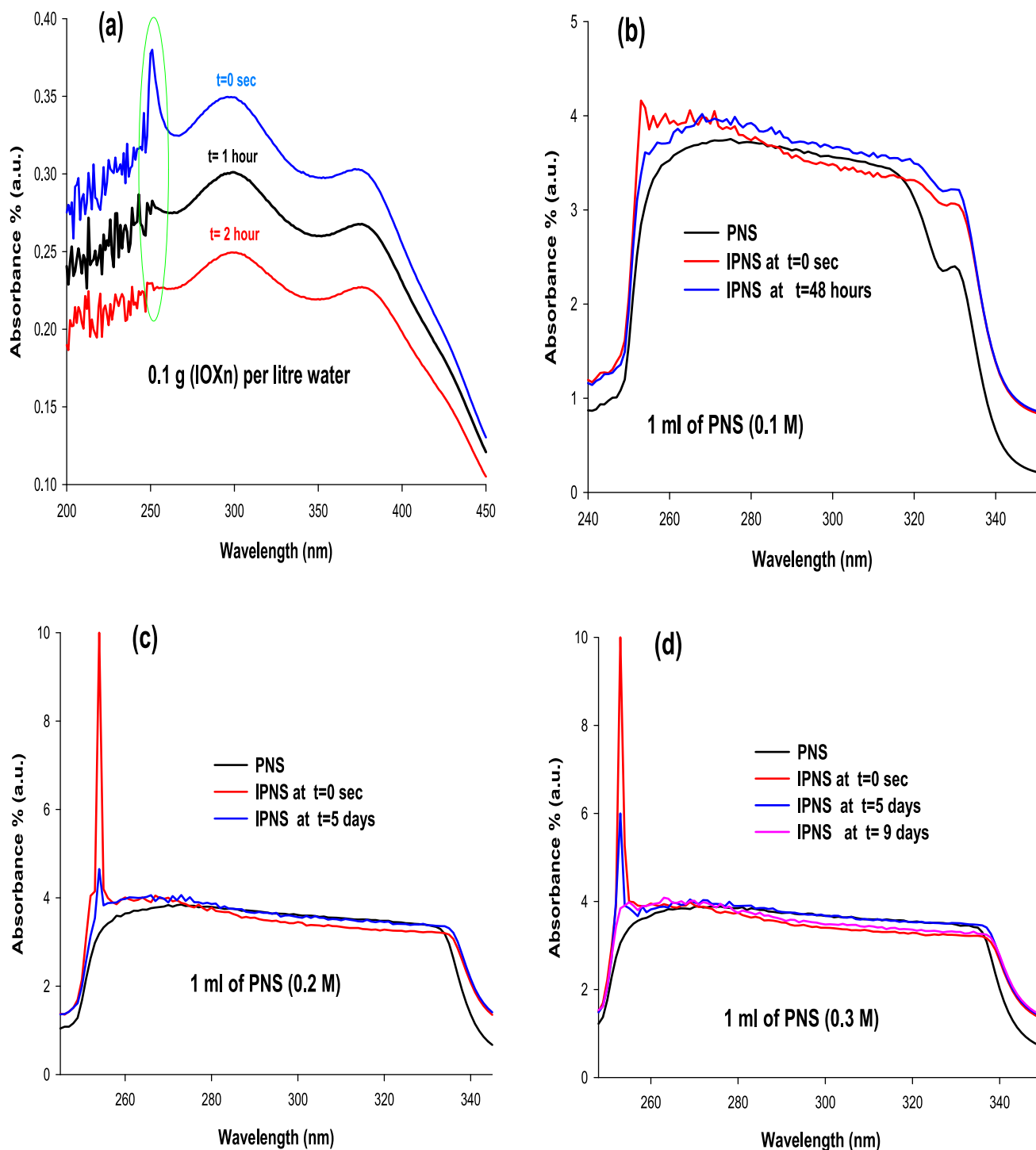


Fig. 10. UV-vis spectra of dispersed IOXn in distilled water (a) alone, (b) with 0.1 M of PNS, (c) with 0.2 M of PNS, and (d) with 0.3 M of PNS.

endothermic peaks at 200 and 350 °C may be seen on the DTA curve above [27] and an exothermic peak at around 547 °C, confirms this. Between 400 and 600 °C, IPNS loses around 20 % of its weight due to the SO_3 . However, the magnetite is stable and can be found at temperatures higher than 600°Celsius in the sample. The thermal stability of PNS is generally high, as is the thermal stability of inorganic IOXn. As a result, the IPNS could be useful in thermal applications.

3.11. UV-Vis spectroscopy analysis

UV-vis spectral analysis is used to determine the stability of IOXn in an aqueous colloidal solution. The distinctive surface of the black color IOXn is produced from ferrous chloride and ferric chloride with a molar ratio of 1:1. A Plasmon absorption band at 250 nm was observed at a concentration 0.1 g/liter. The ultrasonic sonicator effectively disseminated the nanoparticles. UV Cuvette

was filled with the solution. UV spectroscopy was used to scan and measure the stability of nanoparticles. If the scanning range is too high, too low, or too wide in the spectrum, the solution should be diluted or reconstituted, and the wavelength and absorbance should be adjusted. Fig. 10a shows the UV–Vis spectra of IOXn. Pure IOXn has a high peak intensity at a time ($t = 0$), indicating that the nanoparticles are well disseminated and stable in the solution bulk. After an hour ($t = 1$ h), the peak intensity disappeared, indicating that the nanoparticles had precipitated. The PNS ultraviolet spectrum has two distinct absorption bands between 250 and 320 nm, attributed to π - π^* electron transitions inside the benzenoid ring ($C=C - C=C$) comprising two double bonds conjugated systems that exist in PNS. The ultraviolet spectra of PNS molecules reflect the properties of the chromophores groups and assistant chromophores group [28,29]. Fig. 10 (b–d) demonstrate the absorbance and wavelength of PNS at concentrations of 0.1, 0.2, and 0.3 M. With increased PNS concentration, the absorption band of IOXn and IPNS appear at the same wavelength area (253 nm). With increasing PNS concentration, the intensity of the IOXn absorbance band grows until it reaches a constant value. The maximum absorbance bands are (0.38), (4.2), (10), and (10) for IOXn without additive and IOXn with (0.1), (0.2), and (0.3 M) of PNS, respectively. The strength of the IOXn's absorbance band changed over time, which coincides with IOXn dispersion stability in bulk solution. The well-dispersed IOXn produces absorbance at 0.38 (at $\lambda = 250$ nm) without PNS. The absorption intensity disappeared after 1 h. As a result, all IOXn have settled. In the case of adding 0.1, 0.2, and 0.3 M of PNS to IOXn, the absorbance intensities of IPNS with different concentrations were reduced after 3, 5, and 9 days respectively.

4. Conclusion

In this study, sodium poly (naphthalene formaldehyde) sulfonate was successfully used to coat iron oxide nanoparticles, making them easier to manufacture for stable dispersion in an aqueous solution. The structure and dispersion stability of the coating process are characterized and confirmed by every analysis. The PNS, which has a high average molecular weight (26561 Dalton) and high thermal stability, was successfully synthesized. FT-IR analysis confirms the characteristic peaks on the surface of iron oxide. At pH = 8, the zeta potential shifted from -13 to -23 mv. Iron oxide nanoparticles lost their crystallinity and became amorphous forms. The average particle size was reduced from 138 to 32 nm. The Magnetization Saturation (M_s) slightly dropped from 62.2 to (37.8) emu/g. The specific surface area was reduced from 35.049 to 30.686 m^2/g . The morphological images demonstrate that nanoparticles disperse without aggregating. The contact angle of water was slightly reduced by adding IPNS. However, the addition of IPNS had no impact on the surface tension. UV–vis analysis explains the high dispersion stability with increased concentration of PNS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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